Ganit

data speaks

MAX Life Insurance

Data Migration – On-Premises to Google Cloud Platform

30th October 2025

Migrating MAX Life's Core Data Platform to GCP: Scalability, Compliance & Analytics at Scale

Problem Statement & Challenges

MAX Life's on-premises data infrastructure housed critical insurance, actuarial, and customer data across siloed systems. Aging hardware, data access bottlenecks, and IRDAI compliance demands required a move to a modern, scalable, and auditable cloud environment.

On-Prem Scalability Limits

Inability to scale compute & storage for growing policy volumes

Data Siloes & Latency

Disconnected actuarial, CRM, and claims systems with stale data

IRDAI Compliance Gaps

Audit trails and data residency requirements unmet by legacy stack

Analytics Bottlenecks

No real-time reporting; heavy batch jobs delayed business decisions

Solution Proposed by Ganit

End-to-end GCP migration using BigQuery, Dataproc, and Cloud Composer with IRDAI-compliant security architecture.

Solution Overview

Lift & Shift + Re-platform

On-prem Oracle/SQL Server DBs migrated to Cloud SQL & BigQuery using Datastream CDC.

GCP Data Lake Architecture

Landed raw, processed, and curated zones on GCS; orchestrated via Cloud Composer (Airflow).

Compliance & Governance

GCP Services Used

- BigQuery
- Datastream
- Cloud SQL
- Data Catalog
- Cloud Composer
- GCS
- Dataproc
- VPC-SC / CMEK / DLP

Business Value Achieved

- Migrated 200+ tables and 50+ data pipelines with zero downtime, covering policy, claims, and actuarial data.
- Achieved full IRDAI compliance with automated audit logging, CMEK encryption, and regional data residency on GCP.
- Real-time analytics via BigQuery enabled 3× faster actuarial reporting and reduced month-end close time by 40%.
- Infrastructure costs reduced by 22% through autoscaling Dataproc clusters and elimination of idle on-prem hardware.
- Unified data platform enabled cross-functional dashboards in Looker Studio, accelerating business insights delivery.

Legacy On-Premises Infrastructure Couldn't Support MAX Life's Growth Ambitions

Problem Statement

MAX Life Insurance operated its core data infrastructure on-premises — spanning Oracle databases, SQL Server data marts, and legacy ETL pipelines processing millions of daily insurance transactions. As policyholder growth accelerated and IRDAI introduced stringent data governance mandates, the platform struggled with scalability, compliance, and the inability to deliver real-time analytics to business teams. The objective was to modernize the data estate with a cloud-native GCP foundation.

Aging On-Prem Hardware

Oracle and SQL Server instances on EOL hardware causing frequent outages and performance degradation during peak policy renewal periods.

Data Silos & Redundancy

Separate systems for policy, claims, CRM, and actuarial data with no unified layer — leading to inconsistent MIS reports across business units.

IRDAI Compliance Gaps

Inability to demonstrate full audit trails, data residency within India, and encryption-at-rest as mandated by IRDAI circular guidelines.

Analytics Bottlenecks

Month-end actuarial runs taking 12+ hours; no self-service BI capability; leadership reliant on weekly static reports for business decisions.

Ganit Designed a Cloud-Native GCP Data Platform for MAX Life's Migration

Solution Overview

Cloud Migration & Modernization

Migrated on-premises Oracle and SQL Server sources to Cloud SQL and BigQuery using Datastream CDC for zero-downtime cutover. Initial full-load via Storage Transfer Service followed by continuous incremental sync.

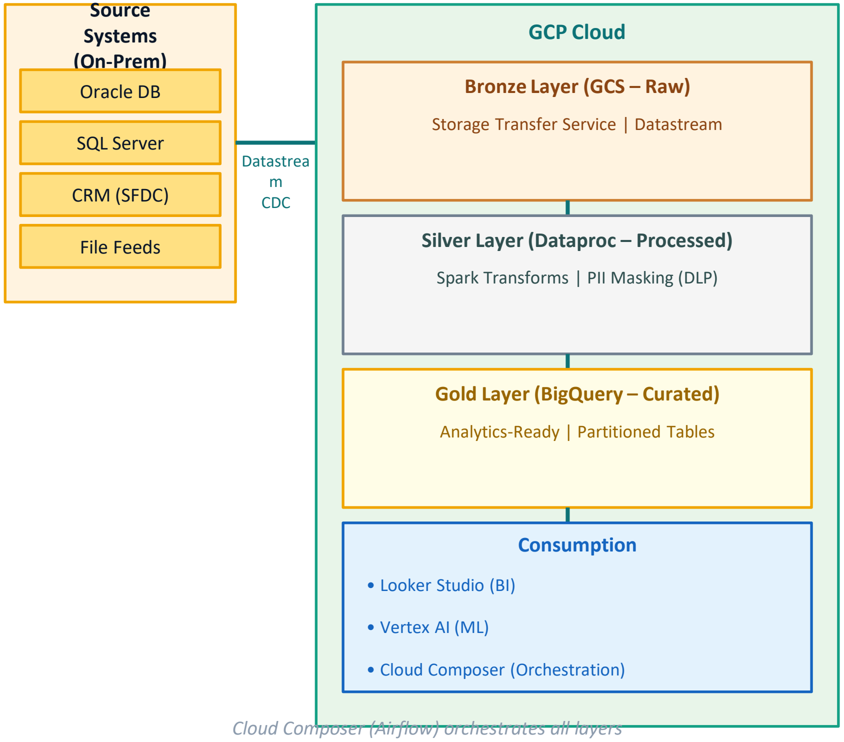
GCP Medallion Data Lake

Established Bronze (raw GCS), Silver (Dataproc Spark transforms), and Gold (BigQuery curated datasets) layers. Orchestrated by Cloud Composer (Airflow 2.x) with dependency-aware DAG scheduling.

Compliance & Data Governance

VPC Service Controls enforced data perimeter. CMEK (Cloud KMS) for encryption at rest. Cloud DLP for PII detection & masking. Google Cloud Data Catalog for metadata lineage — meeting IRDAI mandates.

Data Architecture



Ganit Delivered Measurable Business Value for MAX Life

Outcome: Delivered a fully cloud-native, IRDAI-compliant, and analytics-ready data platform on GCP that consolidated MAX Life's policy, claims, and actuarial data into a single source of truth — enabling real-time decisions and significant cost reductions.

Operational Efficiency

40%

Faster month-end close

Cloud Composer automation eliminated manual ETL hand-offs. AWS-style autoscaling replaced static on-prem provisioning.

Cost Optimization

22%

Reduction in infra cost

Autoscaling Dataproc clusters and BigQuery on-demand pricing replaced always-on on-prem hardware.

Scalability & Reliability

99.9%

Platform uptime SLA

GCP multi-region architecture with failover eliminated single-point-of-failure issues that caused past outages.

Compliance & Governance

100%

IRDAI audit readiness

CMEK, VPC-SC, DLP masking, and Data Catalog lineage deliver end-to-end auditability mandated by IRDAI.